Correlation-Aware Signal Assessment for Systematic Strategies

A Companion Paper

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Abstract

Raw trading signals produced by forecasting models encode directional views about asset returns, but they carry no information about portfolio correlation structure. Sizing positions in direct proportion to signal strength concentrates risk in correlated clusters and understates true portfolio exposure. This paper describes the problem of signal assessment in the presence of non-trivial asset correlations, explains why naive proportional sizing and standard production optimisers both make it difficult to evaluate pure signal quality, and presents the analytical framework implemented in the **Basanos** library. Basanos treats position sizing as an inverse correlation problem—solving the normalised linear system $\mathbf{C}\,\mathbf{x} = \boldsymbol{\mu}$ per timestamp—and couples this with a suite of portfolio-level and optimizer-level diagnostics that let a practitioner measure how much of the realised performance is attributable to the signal itself.

Contents

1 Introduction **3**

2 The Signal Assessment Problem **3**
2.1 Notation . . . 3
2.2 Signal-Proportional Sizing and Its Limitations . . . 4
2.3 Impact of Production Constraints . . . 4

3 The Basanos Approach **4**
3.1 Overview . . . 4
3.2 Volatility Adjustment . . . 5
3.3 EWMA Correlation Estimation . . . 5
3.4 Shrinkage Toward the Identity Matrix . . . 6
3.5 The Normalised Linear Solve . . . 6
3.6 Shrinkage as a Sensitivity Analysis . . . 7

4 Factor Risk Models **8**
4.1 The Factor Model Decomposition . . . 8
4.2 Latent Factors via the Singular Value Decomposition . . . 8
4.3 Efficient Inversion via the Sherman–Morrison–Woodbury Identity . . . 9
4.4 A Sliding Window with k Factors . . . 10

5 Available Analytics **11**
5.1 Optimizer-Level Diagnostics . . . 11
5.1.1 Condition Number . . . 11
5.1.2 Effective Rank . . . 11
5.1.3 Solver Residual . . . 11
5.1.4 Signal Utilisation . . . 12
5.1.5 Position Leverage . . . 12
5.1.6 Risk Position . . . 12
5.2 Signal Quality Metrics . . . 12
5.2.1 Information Coefficient (IC) . . . 12
5.2.2 Rank IC . . . 12
5.2.3 Summary Statistics . . . 13
5.3 Portfolio-Level Analytics . . . 13
5.3.1 P&L and NAV . . . 13
5.3.2 Risk and Drawdown . . . 13
5.3.3 Performance Attribution . . . 13
5.3.4 Turnover . . . 13
5.4 Statistical Metrics . . . 14
5.5 Visualisations and Reports . . . 15

6 Usage Example **15**
6.1 End-to-End Example . . . 15

7 Computational Characteristics **16**

8 Conclusion **17**

A Configuration Parameters **17**

1 Introduction

Systematic trading strategies produce a signal vector $\boldsymbol{\mu}_t = (\mu_{1,t}, \ldots, \mu_{n,t}) \in \mathbb{R}^n$ at each decision time t, where $\mu_{i,t}$ encodes the model's view on the expected excess return of asset i. The immediate question facing any practitioner is: *does this signal contain genuine predictive information, and if so, how much?*

At first glance the answer seems straightforward—run the signal on historical data, record the P&L, and compute the Sharpe ratio. In practice two structural complications make this naive assessment misleading:

1. **Signal-proportional sizing ignores correlations.** If two assets are strongly correlated and both carry a positive signal, placing a large notional in each concentrates risk rather than diversifying it. The realised Sharpe depends on how the signal interacts with the correlation structure, not on the signal alone.

2. **Production optimisers introduce confounders.** A fully constrained Markowitz optimiser—with turnover limits, sector caps, leverage constraints, and factor-neutrality targets—will bend positions away from what the signal implies. P&L measured through such a system reflects the interaction between the signal and all those constraints, making it hard to isolate the signal's intrinsic value.

Basanos addresses both issues through a single, intentionally minimal framework. It solves a normalised inverse-correlation system to map the signal into positions that account for portfolio geometry, while deliberately avoiding hard constraints so the signal can express itself cleanly. This makes it a natural *first hurdle*: a signal that cannot generate a reasonable risk-adjusted return through this framework is unlikely to survive the added friction of a production system.

The paper is structured as follows. Section 2 formalises the signal assessment problem. Section 3 develops the mathematical pipeline—volatility adjustment, EWMA correlation estimation, shrinkage, and the normalised linear solve—and closes with a shrinkage sweep that quantifies what correlation adjustment adds over a proportional baseline. Section 4 presents factor risk models as an alternative covariance parameterisation, including latent-factor extraction via SVD, efficient inversion via the Woodbury identity, and a sliding-window estimator. Section 5 documents optimizer-level diagnostics and portfolio-level analytics. Section 6 illustrates the complete workflow with a code example. Section 7 summarises computational complexity and practical limits. Section 8 concludes.

2 The Signal Assessment Problem

2.1 Notation

Let n denote the number of assets and T the number of decision timestamps. We write:

$$\boldsymbol{\mu}_t \in \mathbb{R}^n \qquad \text{expected-return signal at time } t, \tag{1}$$

$$\mathbf{r}_t \in \mathbb{R}^n \qquad \text{vector of asset returns at time } t, \tag{2}$$

$$\mathbf{C}_t \in \mathbb{R}^{n \times n} \qquad \text{correlation matrix at time } t, \tag{3}$$

$$\boldsymbol{\sigma}_t \in \mathbb{R}^n_{>0} \qquad \text{per-asset return volatilities at time } t. \tag{4}$$

A portfolio is specified by a cash-position vector $\mathbf{p}_t \in \mathbb{R}^n$, where $p_{i,t}$ is the dollar notional held in asset i at time t.

2.2 Signal-Proportional Sizing and Its Limitations

We now make the interaction between signal direction and correlation structure precise. The simplest sizing rule is *proportional*: set

$$p_{i,t} \propto \mu_{i,t}. \tag{5}$$

This rule is intuitive but problematic. If assets i and j are highly correlated ($C_{ij,t} \approx 1$) and $\mu_{i,t}, \mu_{j,t} > 0$, both receive large positive positions and the portfolio's realised exposure to their common factor is approximately $p_{i,t} + p_{j,t}$. The Sharpe ratio of such a portfolio depends not just on $\|\boldsymbol{\mu}_t\|$ but on the angle between $\boldsymbol{\mu}_t$ and the eigenvectors of $\mathbf{C}_t$.

Concretely, suppose the portfolio return at time t is

$$R_t = \mathbf{p}_t^\top \mathbf{r}_t. \tag{6}$$

Under proportional sizing $\mathbf{p}_t = k\,\boldsymbol{\mu}_t$ for some scalar k,

$$\mathrm{Var}(R_t) = k^2\,\boldsymbol{\mu}_t^\top\,\boldsymbol{\Sigma}_t\,\boldsymbol{\mu}_t, \tag{7}$$

where $\boldsymbol{\Sigma}_t = \mathrm{diag}(\boldsymbol{\sigma}_t)\,\mathbf{C}_t\,\mathrm{diag}(\boldsymbol{\sigma}_t)$ is the covariance matrix. The information ratio is

$$\mathrm{IR} = \frac{k\,\boldsymbol{\mu}_t^\top\boldsymbol{\mu}_t}{\sqrt{k^2\,\boldsymbol{\mu}_t^\top\boldsymbol{\Sigma}_t\,\boldsymbol{\mu}_t}} = \frac{\|\boldsymbol{\mu}_t\|^2}{\sqrt{\boldsymbol{\mu}_t^\top\boldsymbol{\Sigma}_t\,\boldsymbol{\mu}_t}}. \tag{8}$$

When assets are perfectly correlated, $\boldsymbol{\Sigma}_t = \sigma^2\mathbf{1}\mathbf{1}^\top$ and $\boldsymbol{\mu}_t^\top\boldsymbol{\Sigma}_t\,\boldsymbol{\mu}_t = \sigma^2(\mathbf{1}^\top\boldsymbol{\mu}_t)^2$, so the IR is determined almost entirely by the sum of signals rather than by their individual strengths. In other words, the signal's *correlation structure* dominates the measured performance.

2.3 Impact of Production Constraints

A Markowitz optimiser (Markowitz, 1952) with turnover, leverage, sector, and factor constraints solves

$$\max_{\mathbf{p}}\ \boldsymbol{\mu}_t^\top\mathbf{p} - \tfrac{1}{2\lambda}\,\mathbf{p}^\top\boldsymbol{\Sigma}_t\,\mathbf{p} \quad \text{subject to a set of linear constraints.} \tag{9}$$

The solution $\mathbf{p}_t^*$ depends on all constraints simultaneously. Adding a turnover limit, for example, ties today's portfolio to yesterday's, so a signal spike one day will be smoothed over multiple periods. The realised P&L then reflects the interaction between the signal's autocorrelation properties and the turnover budget—not the signal's raw predictive content. Boyd et al. (2024) provide a comprehensive treatment of such a fully constrained production Markowitz solver—covering multi-period objectives, transaction cost models, and factor risk—serving as an authoritative reference for what a production system typically entails.

This entanglement makes it difficult to answer the core question: does the signal have edge? Basanos deliberately avoids hard constraints to give the signal room to express itself, while still handling the correlation geometry correctly. The result is interpretable as the *unconstrained potential* of the signal—an upper bound on what a constrained production system can achieve from the same alpha source.

3 The Basanos Approach

3.1 Overview

Basanos computes cash positions through a three-step pipeline:

1. **Volatility adjustment** — normalise returns and compute per-asset EWMA volatility.

2. **Correlation estimation with shrinkage** — compute the time-varying EWMA correlation matrix and regularise it toward the identity.

3. **Normalised linear solve** — solve $\mathbf{C}_t\,\mathbf{x}_t = \boldsymbol{\mu}_t$ for the normalised risk position vector, then divide by per-asset volatility to obtain cash positions.

3.2 Volatility Adjustment

Let $S_{i,t}$ be the price of asset i at time t. The log return is $r_{i,t} = \ln(S_{i,t}/S_{i,t-1})$. An EWMA volatility estimate with half-life h_σ (corresponding to the configuration parameter `vola`) is computed as

$$\sigma^2_{i,t} = \alpha\, r^2_{i,t} + (1-\alpha)\,\sigma^2_{i,t-1}, \quad \alpha = 1 - e^{-\ln 2/h_\sigma}. \tag{10}$$

The volatility-adjusted, clipped return is then

$$\tilde{r}_{i,t} = \mathrm{clip}\left(\frac{r_{i,t}}{\sigma_{i,t}},\ -c,\ c\right), \tag{11}$$

where c is the clipping threshold (`cfg.clip`). Clipping limits the influence of outliers on the correlation estimate and prevents extreme positions when volatility is temporarily suppressed.

3.3 EWMA Correlation Estimation

The EWMA correlation matrix $\hat{\mathbf{C}}_t$ with half-life h_ρ (`cfg.corr`, required $\geq h_\sigma$) is computed component-wise. Because $\hat{C}_{ij,t} = \hat{C}_{ji,t}$, it is sufficient to update only the $n(n+1)/2$ pairs with $j \geq i$; the lower triangle is then filled by symmetry. For each such pair:

$$E_{i,t} = \alpha_\rho\,\tilde{r}_{i,t} + (1-\alpha_\rho)\,E_{i,t-1}, \tag{12}$$

$$E_{ij,t} = \alpha_\rho\,\tilde{r}_{i,t}\tilde{r}_{j,t} + (1-\alpha_\rho)\,E_{ij,t-1}, \tag{13}$$

$$E_{i^2,t} = \alpha_\rho\,\tilde{r}^2_{i,t} + (1-\alpha_\rho)\,E_{i^2,t-1}, \tag{14}$$

where $\alpha_\rho = 1 - e^{-\ln 2/h_\rho}$. The correlation is

$$\hat{C}_{ij,t} = \frac{E_{ij,t} - E_{i,t}\,E_{j,t}}{\sqrt{(E_{i^2,t} - E^2_{i,t})(E_{j^2,t} - E^2_{j,t})}}. \tag{15}$$

The five running moments (E_i, E_j, E_{i^2}, E_{j^2}, E_{ij}) are advanced in parallel across all $n(n+1)/2$ upper-triangular pairs using `scipy.signal.lfilter`, making the step $O(Tn^2)$ in both time and memory (Hunter, 1986).

Outer-product interpretation. Stacking the volatility-adjusted returns into a vector $\tilde{\mathbf{r}}_t = (\tilde{r}_{1,t}, \ldots, \tilde{r}_{n,t})^\top$, the EWMA cross-moment matrix can be written compactly as

$$\mathbf{M}_t = \alpha_\rho\,\tilde{\mathbf{r}}_t\tilde{\mathbf{r}}_t^\top + (1-\alpha_\rho)\,\mathbf{M}_{t-1}, \tag{16}$$

i.e. the exponentially weighted sum of rank-1 outer products $\tilde{\mathbf{r}}_t\tilde{\mathbf{r}}_t^\top$. Each outer product contributes all $n(n+1)/2$ upper-triangular cross-moment estimates simultaneously with a single pass over the data. The correlation matrix $\hat{\mathbf{C}}_t$ is then obtained by mean-centering $\mathbf{M}_t$ and normalising each entry by the corresponding diagonal elements, as in the component-wise update above. This outer-product structure makes the construction equivalent to an exponentially weighted sample covariance of the vol-adjusted returns, and is the natural generalisation of a standard EWMA covariance estimator to the correlation scale.

3.4 Shrinkage Toward the Identity Matrix

Sample correlation matrices are poorly conditioned when n is large relative to T: the Marchenko–Pastur law (Marčenko and Pastur, 1967) shows that extreme eigenvalues of the sample matrix are severely biased. Small eigenvalues are deflated, large ones inflated, and directly inverting a poorly conditioned matrix amplifies noise.

Basanos applies convex linear shrinkage (Ledoit and Wolf, 2004) toward the identity matrix:

$$\tilde{\mathbf{C}}_t = \lambda\, \hat{\mathbf{C}}_t + (1-\lambda)\, \mathbf{I}_n, \tag{17}$$

where $\lambda \in [0,1]$ is the shrinkage retention weight (`cfg.shrink`). Two corner cases are instructive:

- $\lambda = 1$ — the raw EWMA matrix is used without regularisation. This is numerically unstable when n/T is large.
- $\lambda = 0$ — $\tilde{\mathbf{C}}_t = \mathbf{I}_n$. The system $\mathbf{I}\,\mathbf{x} = \boldsymbol{\mu}$ has the trivial solution $\mathbf{x} = \boldsymbol{\mu}$, recovering signal-proportional sizing (Eq. (5)). This is the uncorrelated baseline and matches what a Markowitz solver produces when all assets are treated as independent.

A useful starting heuristic is

$$\lambda \approx 1 - \frac{n}{2T}, \tag{18}$$

which approximates the Ledoit–Wolf optimal intensity (Ledoit and Wolf, 2004). Table 1 summarises the recommended ranges.

Table 1: Shrinkage retention weight λ as a function of the concentration ratio n/T.

Regime	n/T	**Suggested** λ
Many assets, short lookback	> 0.5	0.3–0.5
Moderate assets and lookback	0.1–0.5	0.5–0.7
Few assets, long lookback	< 0.1	0.7–0.9

3.5 The Normalised Linear Solve

For each timestamp t, Basanos solves the linear system

$$\tilde{\mathbf{C}}_t\, \mathbf{x}_t = \boldsymbol{\mu}_t, \tag{19}$$

where $\tilde{\mathbf{C}}_t$ is the shrunk correlation matrix from Eq. (17). The solution is computed via Cholesky decomposition (falling back to LU if the matrix is not numerically positive definite).

Scale invariance. The raw solution $\mathbf{x}_t = \tilde{\mathbf{C}}_t^{-1}\boldsymbol{\mu}_t$ scales linearly with $\|\boldsymbol{\mu}_t\|$, which would make position sizes depend on the arbitrary amplitude of the signal. To achieve scale invariance, positions are normalised by the inverse-matrix norm of $\boldsymbol{\mu}_t$:

$$\nu_t = \sqrt{\boldsymbol{\mu}_t^\top \tilde{\mathbf{C}}_t^{-1} \boldsymbol{\mu}_t}, \tag{20}$$

giving the normalised risk position

$$\hat{\mathbf{x}}_t = \frac{\tilde{\mathbf{C}}_t^{-1}\boldsymbol{\mu}_t}{\nu_t}. \tag{21}$$

Doubling $\boldsymbol{\mu}_t$ leaves $\hat{\mathbf{x}}_t$ unchanged because both numerator and denominator scale identically. This separates signal direction from position magnitude, which is controlled by `cfg.position_scale`.

Cash positions. Converting risk positions to dollar notionals requires dividing by per-asset volatility:

$$p_{i,t} = \frac{\hat{x}_{i,t}}{\sigma_{i,t}} \times \delta, \tag{22}$$

where δ is a position-scale constant (`cfg.position_scale`, default 10^6).

Remark 1. *The normalised risk position in Eq. (21) lives on the ellipsoid* $\{\mathbf{x} : \mathbf{x}^\top \tilde{\mathbf{C}}_t \mathbf{x} = 1\}$. *Geometrically, it is the direction in signal space that maximises the portfolio's information ratio under the given correlation structure. This is the Markowitz efficient direction in the unconstrained, full-information limit. Figure 1 illustrates this for* $n = 2$ *assets.*

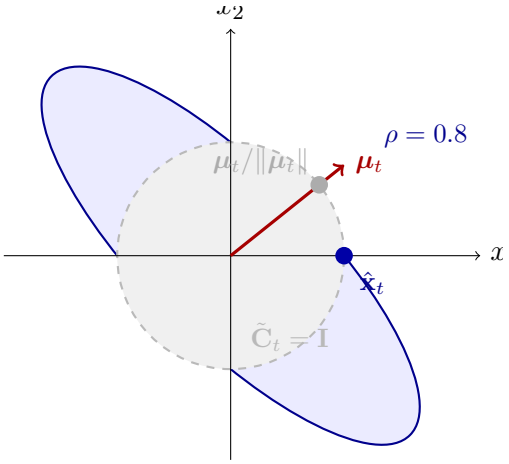


Figure 1: Geometric interpretation for $n = 2$ assets with signal $\boldsymbol{\mu}_t = (1,\, 0.8)^\top$ (red arrow). **Grey circle**: feasible set under no correlation adjustment ($\tilde{\mathbf{C}}_t = \mathbf{I}$); the proportional solution $\boldsymbol{\mu}_t/\|\boldsymbol{\mu}_t\|$ (grey dot) lies in the signal direction. **Blue ellipse**: feasible set $\{\mathbf{x} : \mathbf{x}^\top \tilde{\mathbf{C}}_t \mathbf{x} = 1\}$ for $\rho = 0.8$; the correlation-adjusted solution $\hat{\mathbf{x}}_t = (1,\, 0)^\top$ (blue dot) eliminates the position in asset 2, whose contribution to portfolio risk is already captured through the highly correlated asset 1. The ellipse is compressed in the correlated direction $(1, 1)^\top$ and stretched in the anti-correlated direction $(1, -1)^\top$.

3.6 Shrinkage as a Sensitivity Analysis

Because $\lambda = 0$ recovers signal-proportional sizing and $\lambda = 1$ uses the raw EWMA matrix, sweeping λ between these extremes provides a direct measurement of how much value correlation adjustment adds. Basanos exposes this as the method `engine.sharpe_at_shrink(λ)`, which re-runs the optimiser with a given λ and returns the annualised Sharpe ratio.

A complementary baseline is `engine.naive_sharpe`, which computes the Sharpe ratio when each asset is positioned proportionally to its signal with no correlation adjustment (equivalent to $\lambda = 0$). Together, `naive_sharpe`, `sharpe_at_shrink(0.0)`, and `sharpe_at_shrink(cfg.shrink)` form a three-point diagnostic that separates raw signal skill from the value added by correlation routing.

A typical sweep over $\lambda \in [0, 1]$ with fixed prices and signals yields a Sharpe curve that either:

1. **Peaks at an interior** λ: correlation adjustment adds value, and the optimal intensity is data-driven.

2. **Monotonically decreases from** $\lambda = 0$: the signal is already well-diversified and the EWMA correlation estimate introduces estimation noise.

3. **Monotonically increases toward $\lambda = 1$**: a strong correlation structure exists in the data that the signal does not yet exploit.

This diagnostic is a fast, parameter-free sanity check before committing to a production deployment.

4 Factor Risk Models

The EWMA-plus-shrinkage framework described in Section 3 estimates a full $n \times n$ correlation matrix at every timestamp and regularises it toward the identity. An alternative and complementary approach is to posit a *factor structure* for the covariance matrix: the returns of n assets are driven by a small number $k \ll n$ of common factors plus independent idiosyncratic noise. This section develops that structure, shows how latent factors can be extracted from data via the Singular Value Decomposition (SVD), derives an efficient solve via the Sherman–Morrison–Woodbury identity, and describes a sliding-window estimator parametrised by the pair (n, k).

4.1 The Factor Model Decomposition

A factor risk model decomposes the $n \times n$ covariance matrix as

$$\mathbf{\Sigma} = \mathbf{B}\mathbf{F}\mathbf{B}^\top + \mathbf{D}, \tag{23}$$

where

- $\mathbf{B} \in \mathbb{R}^{n \times k}$ is the *factor loading matrix*: column j gives the sensitivity of each asset to factor j;
- $\mathbf{F} \in \mathbb{R}^{k \times k}$ is the *factor covariance matrix* (positive definite), capturing how the k factors co-vary with one another; and
- $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_n)$ with $d_i > 0$ is the *idiosyncratic risk matrix*, capturing the asset-specific variance unexplained by the common factors.

The central assumption is $k \ll n$: the dominant systematic sources of risk—market direction, sector exposures, style tilts, and so on—are captured by a handful of factors, while the idiosyncratic component $\mathbf{D}$ is, by construction, uncorrelated across assets. Because $k \ll n$, the model reduces the number of free parameters from $O(n^2)$ to $O(nk)$, making estimation far more tractable when the observation count T is small relative to n.

Classic *explicit* factor models specify $\mathbf{B}$ and $\mathbf{F}$ using economic theory or commercial risk-factor vendors, such as the Fama–French three-factor model (Fama and French, 1993). In the purely data-driven setting explored here the factors are *latent* and must be inferred from returns.

Remark 2. *When $\mathbf{F} = \mathbf{I}_k$ and the columns of $\mathbf{B}$ are orthonormal, Eq. (23) reduces to a principal-component decomposition of $\mathbf{\Sigma}$: each factor corresponds to a principal direction and its loading magnitude equals the corresponding eigenvalue.*

4.2 Latent Factors via the Singular Value Decomposition

The most natural way to extract latent factors from a return matrix is the Singular Value Decomposition (Golub and Van Loan, 1996). Let $\mathbf{R} \in \mathbb{R}^{T \times n}$ be the matrix whose (t, i) entry is the volatility-adjusted return $\tilde{r}_{i,t}$. Its thin SVD is

$$\mathbf{R} = \mathbf{U}\mathbf{\Sigma}_{\mathrm{sv}}\mathbf{V}^\top, \tag{24}$$

where $\mathbf{U} \in \mathbb{R}^{T\times n}$ has orthonormal columns ($\mathbf{U}^\top\mathbf{U} = \mathbf{I}_n$), $\mathbf{\Sigma}_{\text{sv}} = \text{diag}(\sigma_1, \ldots, \sigma_n)$ with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$, and $\mathbf{V} \in \mathbb{R}^{n\times n}$ is orthogonal. The sample correlation matrix (assuming unit-variance columns) satisfies

$$\hat{\mathbf{C}} = \frac{1}{T}\mathbf{R}^\top\mathbf{R} = \frac{1}{T}\mathbf{V}\mathbf{\Sigma}_{\text{sv}}^2\mathbf{V}^\top. \tag{25}$$

The columns $\mathbf{v}_j$ of $\mathbf{V}$ are the *latent factor directions* in asset space: $\mathbf{v}_1$ points in the direction of maximum variance, $\mathbf{v}_2$ in the direction of maximum residual variance orthogonal to $\mathbf{v}_1$, and so on. The scalar σ_j^2/T is the variance explained by factor j.

The Eckart–Young–Mirsky theorem (Eckart and Young, 1936) guarantees that retaining only the top-k singular triplets gives the best rank-k approximation to $\mathbf{R}$ (and to $\hat{\mathbf{C}}$) in both the Frobenius and spectral norms:

$$\mathbf{R}_k = \mathbf{U}_k\mathbf{\Sigma}_k\mathbf{V}_k^\top, \qquad \hat{\mathbf{C}}^{(k)} = \frac{1}{T}\mathbf{V}_k\mathbf{\Sigma}_k^2\mathbf{V}_k^\top + \hat{\mathbf{D}}, \tag{26}$$

where $\mathbf{U}_k \in \mathbb{R}^{T\times k}$, $\mathbf{\Sigma}_k = \text{diag}(\sigma_1, \ldots, \sigma_k)$, $\mathbf{V}_k \in \mathbb{R}^{n\times k}$, and $\hat{\mathbf{D}}$ is a diagonal correction chosen so that $\hat{\mathbf{C}}^{(k)}$ has unit diagonal (i.e., is a valid correlation matrix).

Identifying $\mathbf{B} = \mathbf{V}_k$ and $\mathbf{F} = \mathbf{\Sigma}_k^2/T$ establishes the correspondence with Eq. (23): the latent factors are the right singular vectors, the factor covariance is diagonal with entries σ_j^2/T, and the idiosyncratic matrix $\hat{\mathbf{D}}$ absorbs the residual variance.

Choosing k. A common heuristic is to retain enough components to explain a target fraction θ of total variance:

$$k^* = \min\left\{k : \frac{\sum_{j=1}^k \sigma_j^2}{\sum_{j=1}^n \sigma_j^2} \geq \theta\right\}. \tag{27}$$

Alternatively, the Marchenko–Pastur bulk edge (Marčenko and Pastur, 1967) provides a noise-floor threshold: retain only singular values exceeding $\sigma_{\max}^{\text{MP}} = \sigma_{\text{noise}}\sqrt{(1+\sqrt{n/T})^2}$, where σ_{noise} is an estimate of the noise level. Both criteria typically select k in the range 3–20 for equity universes with daily data.

4.3 Efficient Inversion via the Sherman–Morrison–Woodbury Identity

For large n, the direct solve $\hat{\mathbf{C}}^{(k)}\,\mathbf{x} = \boldsymbol{\mu}$ is expensive because forming and factorising the full $n \times n$ matrix costs $O(n^3)$. The factor structure permits a much cheaper path via the *Woodbury matrix identity* (Woodbury, 1950):

$$(\mathbf{D} + \mathbf{B}\mathbf{F}\mathbf{B}^\top)^{-1} = \mathbf{D}^{-1} - \mathbf{D}^{-1}\mathbf{B}\underbrace{(\mathbf{F}^{-1} + \mathbf{B}^\top\mathbf{D}^{-1}\mathbf{B})^{-1}}_{k\times k}\mathbf{B}^\top\mathbf{D}^{-1}. \tag{28}$$

Because $\mathbf{D}$ is diagonal, $\mathbf{D}^{-1}$ costs $O(n)$. The bracketed term is a $k\times k$ matrix whose inversion costs $O(k^3)$, and the surrounding multiplications cost $O(kn)$. The total cost is therefore $O(k^3 + kn)$, which is $O(kn)$ for $k \ll n$—a dramatic improvement over $O(n^3)$ when n is large and k is small.

Remark 3. *For the pure SVD factor model ($\mathbf{F} = \mathbf{\Sigma}_k^2/T$, $\mathbf{B} = \mathbf{V}_k$), the Woodbury formula specialises to*

$$\left(\hat{\mathbf{D}} + \tfrac{1}{T}\mathbf{V}_k\mathbf{\Sigma}_k^2\mathbf{V}_k^\top\right)^{-1} = \hat{\mathbf{D}}^{-1} - \hat{\mathbf{D}}^{-1}\mathbf{V}_k\left(T\mathbf{\Sigma}_k^{-2} + \mathbf{V}_k^\top\hat{\mathbf{D}}^{-1}\mathbf{V}_k\right)^{-1}\mathbf{V}_k^\top\hat{\mathbf{D}}^{-1}. \tag{29}$$

The inner matrix $T\mathbf{\Sigma}_k^{-2} + \mathbf{V}_k^\top\hat{\mathbf{D}}^{-1}\mathbf{V}_k$ is $k \times k$ and positive definite, admitting a stable Cholesky factorisation.

The *Sherman–Morrison formula* is the rank-1 special case ($k = 1$) of the Woodbury identity:

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\,\mathbf{v}^\top\mathbf{A}^{-1}}{1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}}, \tag{30}$$

valid whenever $1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u} \neq 0$. When the factor model has $k = 1$ (a single dominant factor, e.g. the market), the full matrix inverse follows from a single n-vector computation, making the per-timestamp solve $O(n)$.

4.4 A Sliding Window with k Factors

The EWMA correlation estimator of Section 3.3 assigns exponentially decaying weights to all past observations, maintaining a running estimate that is updated at each timestamp—an *infinite-memory* estimator. An alternative is the *sliding window* (or rolling-window) estimator, which restricts attention to a fixed block of the W most recent observations and assigns them equal weight.

Construction. At timestamp t, let $\mathbf{R}_t \in \mathbb{R}^{W\times n}$ denote the submatrix of volatility-adjusted returns for the window $[t-W+1,\, t]$ across all n assets. Its truncated SVD retaining k components is

$$\mathbf{R}_t \approx \mathbf{U}_{k,t}\mathbf{\Sigma}_{k,t}\mathbf{V}_{k,t}^\top, \tag{31}$$

and the factor-based correlation estimate used for the solve at time t is

$$\hat{\mathbf{C}}_t^{(W,k)} = \frac{1}{W}\mathbf{V}_{k,t}\mathbf{\Sigma}_{k,t}^2\mathbf{V}_{k,t}^\top + \hat{\mathbf{D}}_t, \tag{32}$$

with $\hat{\mathbf{D}}_t$ chosen to enforce unit diagonal. The system $\hat{\mathbf{C}}_t^{(W,k)}\,\mathbf{x}_t = \boldsymbol{\mu}_t$ can then be solved efficiently using the Woodbury identity from Section 4.3.

Parameters and trade-offs. The pair (W, k) jointly controls the estimator:

- **Window length W.** Longer windows yield more stable factor directions but adapt slowly to regime changes. A common rule of thumb is $W \geq 2n$ to ensure the sample covariance is well-posed before truncation.
- **Number of factors k.** Small k (e.g. 1–5) captures only the dominant systematic risk (market, sector), giving a smooth, low-variance estimate. Larger k (up to $\min(W, n)$) captures finer correlation structure at the cost of higher estimation noise. The $k = 1$ case recovers the market-factor model; $k = n$ (full rank) recovers the standard sample correlation matrix.

Comparison with EWMA. The EWMA and sliding-window estimators exhibit complementary properties. EWMA assigns exponentially decaying weights, which is well suited to tracking smooth regime transitions; its estimate is never zero-weight and is always influenced by distant past data to some degree. The sliding window assigns uniform weights within the window and zero outside it: this hard truncation makes the estimator more responsive to sudden structural breaks but introduces a discrete jump whenever a large observation enters or leaves the window. Table 2 summarises the comparison.

Table 2: EWMA correlation versus sliding-window factor model.

Property	**EWMA (shrinkage)**	**Sliding window (W, k)**
Observation weights	Exponential decay	Uniform within window
Memory horizon	Infinite (soft)	Exactly W steps (hard)
Free parameters	λ, h_ρ	W, k
Covariance rank	Full n	At most k (plus diagonal)
Per-step solve cost	$O(n^3)$	$O(k^3 + kn)$
Regime responsiveness	Gradual	Sharp at boundary

5 Available Analytics

The library exposes two layers of analytics. The first—optimizer-level diagnostics—is accessed through `BasanosEngine` and introspects the numerical behaviour of the solve at each timestamp: condition number, effective rank, solver residual, signal utilisation, and leverage. The second—portfolio-level analytics—is accessed through the `Portfolio` object returned by `engine.portfolio` and measures the risk-adjusted performance of the resulting positions.

5.1 Optimizer-Level Diagnostics

The following properties are available on a `BasanosEngine` instance. Each returns a Polars `DataFrame` indexed by timestamp.

5.1.1 Condition Number

The condition number $\kappa(\tilde{\mathbf{C}}_t)$ of the shrunk correlation matrix quantifies numerical sensitivity of the solve:

$$\kappa(\tilde{\mathbf{C}}_t) = \frac{\lambda_{\max}(\tilde{\mathbf{C}}_t)}{\lambda_{\min}(\tilde{\mathbf{C}}_t)}, \tag{33}$$

where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and smallest eigenvalues respectively. A value near n is expected for a healthy, lightly shrunk matrix. Values above 10^{12} trigger a warning and indicate that positions may be numerically unreliable.

5.1.2 Effective Rank

The effective rank measures the true dimensionality of the portfolio by computing the Shannon entropy of the eigenvalue spectrum of $\tilde{\mathbf{C}}_t$:

$$\text{eff-rank}(\tilde{\mathbf{C}}_t) = \exp\left(-\sum_{i=1}^{n} p_i \ln p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \tag{34}$$

where λ_i are the eigenvalues of $\tilde{\mathbf{C}}_t$. A value of n indicates a perfectly uniform spectrum (identity-like matrix, assets effectively independent). A value near 1 indicates a rank-1 matrix dominated by a single common factor. Monitoring effective rank over time reveals whether the portfolio's correlation structure is diversified or concentrated.

5.1.3 Solver Residual

After solving the system in Eq. (19), the Euclidean residual

$$\varepsilon_t = \left\| \tilde{\mathbf{C}}_t \, \mathbf{x}_t - \boldsymbol{\mu}_t \right\|_2 \tag{35}$$

measures numerical accuracy. For a well-posed, well-conditioned system, ε_t should be near machine epsilon ($\approx 10^{-15}$). Large residuals flag near-singular matrices or solver fallback from Cholesky to LU.

5.1.4 Signal Utilisation

For each asset i, signal utilisation measures the fraction of the original signal that survives the correlation filter:

$$u_{i,t} = \frac{(\tilde{\mathbf{C}}_t^{-1}\boldsymbol{\mu}_t)_i}{\mu_{i,t}}. \tag{36}$$

When $\tilde{\mathbf{C}}_t = \mathbf{I}_n$, all assets have $u_{i,t} = 1$. Off-diagonal correlations attenuate some assets ($u_{i,t} < 1$) and may amplify negatively correlated ones ($u_{i,t} > 1$). This diagnostic reveals which assets are most affected by the correlation adjustment and by how much.

5.1.5 Position Leverage

Gross leverage at each timestamp is the L1 norm of the cash-position vector:

$$L_t = \|\mathbf{p}_t\|_1 = \sum_{i=1}^{n} |p_{i,t}|\,. \tag{37}$$

Monitoring leverage over time reveals whether the correlation adjustment amplifies or dampens total exposure relative to the signal-proportional baseline.

5.1.6 Risk Position

The risk position is the unscaled cash position before dividing by per-asset volatility:

$$x_{i,t}^{\text{risk}} = p_{i,t} \times \sigma_{i,t}. \tag{38}$$

It is returned as a `DataFrame` with the same shape as `cash_position`, and is useful for cross-asset comparison on a common volatility-adjusted scale.

5.2 Signal Quality Metrics

The following properties are available on a `BasanosEngine` instance and evaluate the signal's cross-sectional predictive content independently of any portfolio construction. Each IC observation is computed as a cross-sectional statistic over the active assets at each timestamp.

5.2.1 Information Coefficient (IC)

The IC at timestamp t is the Pearson correlation between the signal $\boldsymbol{\mu}_t$ and the one-step-ahead asset returns $\mathbf{r}_{t+1}$:

$$\text{IC}_t = \text{corr}(\boldsymbol{\mu}_t,\, \mathbf{r}_{t+1}). \tag{39}$$

The property `engine.ic` returns a Polars `DataFrame` of per-timestamp IC values.

5.2.2 Rank IC

The Rank IC is the Spearman rank correlation between $\boldsymbol{\mu}_t$ and $\mathbf{r}_{t+1}$. It is more robust to outliers than the Pearson IC and is the standard metric for evaluating signal quality in the quantitative finance literature. Available as `engine.rank_ic`.

5.2.3 Summary Statistics

- `engine.icir` — Information Coefficient Information Ratio: $\text{ICIR} = \bar{\text{IC}}/\hat{\sigma}_{\text{IC}}$, where $\bar{\text{IC}}$ is the mean and $\hat{\sigma}_{\text{IC}}$ the standard deviation of the IC time series.
- `engine.rank_ic_mean` — Mean of the Rank IC series.
- `engine.rank_ic_std` — Standard deviation of the Rank IC series.

5.3 Portfolio-Level Analytics

After positions are computed, a `Portfolio` object provides a comprehensive set of performance analytics. The key properties are summarised below.

5.3.1 P&L and NAV

- **profits** — Per-asset daily cash P&L: $\pi_{i,t} = p_{i,t-1}\,(S_{i,t}/S_{i,t-1} - 1)$.
- **profit** — Aggregate portfolio daily P&L: $\Pi_t = \sum_i \pi_{i,t}$.
- **nav_accumulated** — Cumulative additive NAV: $\text{NAV}_t = \text{AUM} + \sum_{s \le t} \Pi_s$.
- **nav_compounded** — Compounded NAV: $\text{NAV}^c_t = \text{AUM} \prod_{s \le t}(1 + \Pi_s/\text{AUM})$.

5.3.2 Risk and Drawdown

- **highwater** — Running maximum of the accumulated NAV.
- **drawdown** — Distance from the high-water mark: $\text{DD}_t = (\text{HWM}_t - \text{NAV}_t)/\text{HWM}_t$.
- **returns** — Daily return series: $r_t = \Pi_t/\text{AUM}$.

5.3.3 Performance Attribution

Following Brinson et al. (1986), Basanos decomposes the portfolio return into two orthogonal components:

Tilt (allocation effect) The return due to time-averaged static weights $\bar{\mathbf{p}} = T^{-1}\sum_t \mathbf{p}_t$. This measures the passive, structural exposure of the portfolio.

Timing (selection effect) The return due to dynamic deviations from the static tilt: $\Delta\mathbf{p}_t = \mathbf{p}_t - \bar{\mathbf{p}}$. This isolates the pure signal-driven, time-varying component of performance.

Together they provide a clean decomposition:

$$\Pi_t = \Pi_t^{\text{tilt}} + \Pi_t^{\text{timing}}. \tag{40}$$

A signal with no edge will show a flat or negative timing NAV even if the tilt NAV is positive (due to incidental factor exposures), which directly answers the question of whether the dynamic component adds value.

5.3.4 Turnover

One-way portfolio turnover at time t is

$$\tau_t = \frac{1}{2}\,\|\mathbf{p}_t - \mathbf{p}_{t-1}\|_1\,. \tag{41}$$

High turnover relative to signal magnitude is a warning sign: either the signal is noisy or the correlation matrix is changing rapidly. Turnover also provides an upper bound on the transaction costs that would arise in production.

5.4 Statistical Metrics

The `Stats` class (available as `portfolio.stats`) computes the following metrics on the portfolio returns series. All metrics are available both for the aggregate portfolio and on a per-asset basis.

Risk-adjusted performance.

- **Sharpe ratio** (Sharpe, 1966): $S = \sqrt{252}\,\bar{r}/\hat{\sigma}$, where $\bar{r}$ is the mean daily return and $\hat{\sigma}$ is the sample standard deviation.
- **Calmar ratio**: annualised return divided by maximum drawdown.
- **Recovery factor**: total return divided by maximum drawdown.

Risk metrics.

- **Volatility**: annualised standard deviation of daily returns.
- **Value at Risk (VaR)**: parametric VaR at a given confidence level, computed from the sample mean and standard deviation assuming Gaussian returns.
- **Conditional Value at Risk (CVaR)** (Rockafellar and Uryasev, 2000): expected shortfall beyond the VaR threshold.
- **Maximum drawdown** and **average drawdown**: measures of peak-to-trough loss.
- **Maximum drawdown duration**: longest period spent below the high-water mark.

Distributional properties.

- Mean, skewness, excess kurtosis.
- Best and worst single-day returns.

Win/loss analysis.

- Win rate: fraction of days with positive P&L.
- Monthly win rate: fraction of calendar months with positive P&L.
- Profit factor: gross profit divided by gross loss.
- Payoff ratio: average win divided by average loss.
- Average win and average loss.

Relative performance.

- Up-capture and down-capture ratios versus a user-supplied benchmark.
- Worst k-period windows for stress testing.

Rolling metrics.

- Rolling Sharpe ratio and rolling volatility time series, both with configurable window sizes. These reveal whether performance is concentrated in a short regime or consistently earned across the history.

5.5 Visualisations and Reports

The `Plots` class (available as `portfolio.plots`) produces interactive Plotly figures:

- **NAV snapshot** (`snapshot()`): cumulative NAV and drawdown dashboard.
- **Lead/lag IR plot** (`lead_lag_ir_plot(start, end)`): Sharpe ratio as a function of position lag, revealing whether the signal has predictive power ahead of or behind the positions as submitted. A peak at lag 0 is the expected result for a correctly timed signal.
- **Lagged performance** (`lagged_performance_plot(...)`): NAV curves for multiple position lags overlaid for comparison.
- **Rolling Sharpe** (`rolling_sharpe_plot(window)`): time series of rolling Sharpe ratio to assess performance stability.
- **Rolling volatility** (`rolling_volatility_plot(...)`): time series of rolling annualised volatility.
- **Annual Sharpe** (`annual_sharpe_plot()`): bar chart of Sharpe ratio by calendar year, useful for detecting single-year anomalies.
- **Correlation heatmap** (`correlation_heatmap()`): time-averaged asset correlation matrix.

A one-call `Report` object (`portfolio.report`) renders all of the above as a self-contained, dark-themed HTML document that can be served via an API or saved to disk.

6 Usage Example

6.1 End-to-End Example

The following example illustrates the complete workflow from price data and a signal to a performance report.

Listing 1: End-to-end example: optimization and analytics.

```
import numpy as np
import polars as pl
from basanos.math import BasanosConfig, BasanosEngine

# --- Synthetic data ---
n_days = 500
rng = np.random.default_rng(42)
dates = pl.date_range(
    pl.date(2022, 1, 1),
    pl.date(2022, 1, 1) + pl.duration(days=n_days - 1),
    eager=True,
)
prices = pl.DataFrame({
    "date":  dates,
    "AAPL":  100.0 + np.cumsum(rng.normal(0, 1.0, n_days)),
    "GOOGL": 150.0 + np.cumsum(rng.normal(0, 1.2, n_days)),
    "MSFT":  200.0 + np.cumsum(rng.normal(0, 0.9, n_days)),
})
mu = pl.DataFrame({
    "date":  dates,
    "AAPL":  np.tanh(rng.normal(0, 0.5, n_days)),
```

```
    "GOOGL": np.tanh(rng.normal(0, 0.5, n_days)),
    "MSFT":  np.tanh(rng.normal(0, 0.5, n_days)),
})

# --- Configuration ---
cfg = BasanosConfig(
    vola=16,     # EWMA half-life for volatility (days)
    corr=64,     # EWMA half-life for correlations (>= vola)
    clip=3.5,    # Clipping threshold (standard deviations)
    shrink=0.6, # Shrinkage retention weight lambda
    aum=1e6,     # Assets under management
)

# --- Optimization ---
engine    = BasanosEngine(prices=prices, mu=mu, cfg=cfg)
positions = engine.cash_position       # Optimized cash positions
status    = engine.position_status     # Per-row reason code (warmup
    /zero_signal/degenerate/valid)

# --- Optimizer diagnostics ---
kappa     = engine.condition_number    # Matrix condition over time
eff_rank = engine.effective_rank       # Effective rank over time
residual = engine.solver_residual      # Solve residual over time
utilisation = engine.signal_utilisation  # Per-asset signal
    utilisation

# --- Signal quality metrics ---
ic_series = engine.ic                  # Per-timestamp Pearson IC
rank_ic   = engine.rank_ic             # Per-timestamp Rank IC
print(engine.icir)                     # IC Information Ratio
print(engine.naive_sharpe)             # Sharpe under proportional
    sizing

# --- Portfolio analytics ---
portfolio = engine.portfolio
print(portfolio.stats.sharpe()["returns"])        # Annualised
    Sharpe ratio
print(portfolio.stats.max_drawdown()["returns"])  # Maximum drawdown
decomp = portfolio.tilt_timing_decomp  # Tilt vs. timing NAV curves

# --- Visualisation ---
fig = portfolio.plots.snapshot()           # NAV + drawdown
fig = portfolio.plots.lead_lag_ir_plot(-5, 15)  # Lead/lag Sharpe
portfolio.report.save("output/report")     # Full HTML report
```

7 Computational Characteristics

Let N denote the number of assets and T the number of timestamps.

Memory usage is dominated by storing all correlation matrices: $O(TN^2)$ `float64` values, approximately $8TN^2$ bytes. For $N = 50$, $T = 1260$ (five years of daily data), this is approximately 25 MB.

Practical limits on an 8 GB workstation:

- $N \leq 150$, $T \leq 1260$: seconds.
- $N \leq 250$, $T \leq 2520$: 10–60 seconds, ~12 GB RAM.

Table 3: Computational complexity of the main pipeline steps.

Step	**Complexity**	**Notes**
Volatility adjustment	$O(TN)$	EWMA per asset, linear in both
Correlation estimation	$O(TN^2)$	N^2 pairs; parallelised via `lfilter`
Linear solve (per row)	$O(N^3)$	Cholesky; total $O(TN^3)$
Portfolio analytics	$O(TN)$	Element-wise P&L and NAV

- $N > 500$: exceeds 16 GB RAM.

8 Conclusion

This paper has presented the signal assessment problem that motivates the Basanos library, the correlation-aware linear-solve approach it uses to handle non-trivial portfolio geometry, and the suite of analytics it makes available at both the optimizer and portfolio level.

The key insight is that raw signal-proportional sizing conflates two distinct quantities: (i) the predictive content of the signal and (ii) its interaction with the portfolio correlation structure. By inverting the correlation matrix explicitly—through a normalised solution of $\tilde{\mathbf{C}}_t \, \mathbf{x}_t = \boldsymbol{\mu}_t$—Basanos disentangles these effects. The result is a clean measurement surface on which signal quality can be assessed through standard metrics: Sharpe ratio, drawdown, tilt/timing decomposition, and lead/lag analysis.

The deliberate absence of hard constraints is a feature, not a limitation. It means the realised P&L measures the signal's intrinsic edge rather than its interaction with a production constraint set. A signal that cannot pass this first hurdle is unlikely to survive the additional friction of a full production system.

A Configuration Parameters

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Table 4: `BasanosConfig` parameters and their roles.

Parameter	Description	Default
`vola`	EWMA half-life for volatility (days). Required.	—
`corr`	EWMA half-life for correlation (days, $\geq$ vola). Required.	—
`clip`	Clipping threshold for vol-adjusted returns (> 0). Required.	—
`shrink`	Shrinkage retention weight $\lambda \in [0, 1]$. Required.	—
`aum`	Assets under management for position scaling (> 0). Required.	—
`denom_tol`	Zero threshold for normalisation denominator.	10^{-12}
`position_scale`	Multiplicative scaling applied to risk positions.	10^{6}
`min_corr_denom`	Guard threshold for EWMA correlation denominator.	10^{-14}
`max_nan_fraction`	Maximum tolerated null fraction per asset column.	0.9
`cost_per_unit`	One-way trading cost per unit of position change.	0.0
`max_turnover`	Optional L1 turnover budget per period (cash units).	`None`
`covariance_config`	Covariance mode: `EwmaShrinkConfig` or `SlidingWindowConfig`.	`EwmaShrinkConfig()`

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